CS6493 RAG Project Timeline

Retrieval-Augmented Generation

for Knowledge-Intensive Tasks

Course: CS6493 Natural Language Processing

Semester: Q2 2025-2026

City University of Hong Kong

Group: RAGdoll

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GPU: NVIDIA A100 40GB

Project Timeline Overview

Progress Visualization:

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Phase 1 ======================== Done (3/1 - 3/14)
Phase 2 ======================== Done (3/15 - 3/21)
Phase 3 ======================== In Progress (3/22 - 3/28)
Phase 4 ------------------------ Planned (4/1 - 4/7)
Phase 5 ------------------------ Planned (4/8 - 4/12)
Phase 6 ------------------------ Planned (4/13 - 4/14)
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Phase 1: Project Setup & Environment

Attribute	Detail
Time	Mar 1 - Mar 14
Status	Completed
Week	Week 1-2

Main Tasks

- [x] Determine research topic: Finance + Law dual-domain
- [x] Complete literature review (RAG, FiD, HotpotQA, CUAD, Self-RAG)
- [x] NVIDIA A100 40GB GPU environment setup
- [x] Technology stack selection: LangChain, FAISS, Pyserini

Phase 2: Dataset Collection & Indexing

Attribute	Detail
Time	Mar 15 - Mar 21
Status	Completed
Week	Week 3

Main Tasks

- [x] Collect FinanceBench dataset (~150 QA pairs)
- [x] Collect HotpotQA dev set (~7.4k pairs)
- [x] Collect CUAD dataset (510 contracts, ~13k QA pairs)
- [x] Collect LegalBench-RAG-mini (776 pairs)
- [x] Complete BM25 index construction
- [x] Establish Recall@3 baseline

Dataset Overview

Dataset	Domain	Size	Purpose
FinanceBench	Finance	~150 pairs	SEC 10-K/10-Q QA
HotpotQA	Multi-hop	~7.4k pairs	Multi-hop reasoning
CUAD	Law	~13k pairs	Contract clause extraction

LegalBench-RAG	Law	776 pairs	Legal retrieval benchmark

Phase 3: Retrieval System Implementation

Attribute	Detail
Time	Mar 22 - Mar 28
Status	In Progress
Week	Week 4

Main Tasks

- [x] Contriever integration + FAISS vector indexing
- [ ] Hybrid RRF (Reciprocal Rank Fusion) implementation
- [ ] LangChain RetrievalQA framework setup
- [ ] HotpotQA retrieval precision analysis

Retrieval Methods Comparison

Method	Type	Feature	Use Case
BM25	Sparse	Keyword matching, no GPU	Legal clause matching
Contriever	Dense	Semantic similarity, 768-dim	Cross-language
Hybrid RRF	Mixed	RRF fusion (k=60), Top-k=3	Best overall

Phase 4: Generation Model Integration

Attribute	Detail
Time	Apr 1 - Apr 7
Status	Planned
Week	Week 5

Main Tasks

- [ ] Connect LLaMA-3.1-8B-Instruct
- [ ] Connect Qwen2.5-7B-Instruct
- [ ] Execute 3x2x2 RAG grid experiment
- [ ] Zero-shot vs Instruction-tuned ablation
- [ ] Analyze retrieval precision vs answer quality

3x2x2 Experiment Design

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Retrievers (3): BM25, Contriever, Hybrid RRF
Generators (2): LLaMA-3.1-8B, Qwen2.5-7B
Domains (2): Finance, Law
Total experiments: 3 x 2 x 2 = 12
Metrics: EM, F1, FActScore, Recall@k
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Phase 5: Hallucination Mitigation & Evaluation

Attribute	Detail
Time	Apr 8 - Apr 12
Status	Planned
Week	Week 6

Main Tasks

- [ ] FActScore hallucination rate evaluation
- [ ] LangChain Structured Chain implementation
- [ ] Self-RAG integration comparison
- [ ] Cross-domain performance analysis
- [ ] Human evaluation (50-response sample)

Hallucination Mitigation Strategies

Method	Components	Mechanism
LLM Only	None	Pure parametric, highest hallucination
Vanilla RAG	Retriever + Generator	Standard, no constraints
LangChain	RetrievalQA + Prompt	Forced citation, CoT
Self-RAG	Reflective tokens	Self-critique retrieval

Evaluation Metrics

Metric	Description
EM (Exact Match)	Exact match accuracy
F1 Score	Token-level F1
Recall@k	Retrieval recall rate
FActScore	Factual precision score
Hallucination Rate	Unsupported statements ratio

Phase 6: Final Deliverables

Attribute	Detail
Time	Apr 13 - Apr 14
Status	Planned
Week	Final Week

Main Tasks

- [ ] Final report writing and analysis
- [ ] Presentation slides preparation
- [ ] Code cleanup and documentation
- [ ] End-to-end demo presentation

Task 4 Requirements Mapping

Task 1: Hybrid Generation Interaction

Requirement	Experiment	Phase	Status
2 retrieval methods	BM25 + Contriever + Hybrid	Phase 3-4	Done
2 generation models	LLaMA-3.1-8B + Qwen2.5-7B	Phase 4	Planned
HotpotQA analysis	Recall@k vs Answer F1	Phase 3-4	In Progress
Zero-shot vs Instruct	Qwen base vs Instruct	Phase 4	Planned
Human evaluation	Crowd-sourcing	Phase 5	Planned

Task 2: Multifaceted Mitigation

Requirement	Experiment	Phase	Status
Quantify hallucination	FActScore metrics	Phase 5	Planned
Cross-domain citation	Finance vs Law comparison	Phase 5	Planned
Human evaluation	50-sample stratified	Phase 5	Planned
4 mitigation strategies	LLM/Vanilla/LangChain/Self-RAG	Phase 5	Planned

Experiment Tips

1. Retrieval Document Count: Recommend limiting to 5-9 documents per query to balance performance and computational cost.

2. Current Setting: k=3, can be adjusted based on HotpotQA experiment results.

3. Evaluation Framework: See Appendix for details.

4. Cross-domain Generalization: Pay attention to performance differences between Finance and Law domains.